

Tuesday, February 10

- Thank you for coming to GEMS/ my talk!
- CCMS Applied Math seminar Mondays: 4:15 pm Estella 1021
- CCMS Colloquium Fridays: 11am Davidson Lecture Hall
- No OH Thursday, DM to meet Friday!

Plan

- Code demo
- Empirical Risk Minimization
- Gradient Descent

Linear Regression

$$X \in \mathbb{R}^{n \times d}$$

$$\begin{bmatrix} \vdots \\ \text{--- } x^{(i)T} \text{ ---} \\ \vdots \end{bmatrix}$$

$$y \in \mathbb{R}^n$$

$$\begin{bmatrix} \vdots \\ y^{(i)} \\ \vdots \end{bmatrix}$$

$$\mathcal{L}(w) = \|Xw - y\|_2^2$$

$$\nabla_w \mathcal{L}(w) = 2X^T(Xw - y)$$

Time to compute?



$$\arg \min_w \|Xw - y\|_2^2 = (X^T X)^{-1} X^T y$$

Question: How do we deal with intercept?

Empirical Risk Minimization

Answer to why squared-error

Suppose we believe $y^{(i)} = \langle w^*, x^{(i)} \rangle + \eta^{(i)}$

same as $w^* \cdot x^{(i)}$ but pretty :-)

for noise $\eta \sim \mathcal{N}(0, \sigma^2)$
↑ normal dist

Then $y^{(i)} \sim \mathcal{N}(\langle w^*, x^{(i)} \rangle, \sigma^2)$

$$Pr(y^{(i)}) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y^{(i)} - \langle w, x^{(i)} \rangle)^2}{2\sigma^2}\right)$$

Idea: Find w that maximizes likelihood of observing data

$$\operatorname{argmax}_w Pr(y^{(1)}, \dots, y^{(n)})$$

$$= \operatorname{argmax}_w \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y^{(i)} - \langle w, x^{(i)} \rangle)^2}{2\sigma^2}\right)$$

$$= \operatorname{argmax}_w \exp\left(\sum_{i=1}^n -\frac{(y^{(i)} - \langle w, x^{(i)} \rangle)^2}{2\sigma^2}\right)$$

$$= \operatorname{argmin}_w -\log\left(\exp\left(\sum_{i=1}^n -\frac{(y^{(i)} - \langle w, x^{(i)} \rangle)^2}{2\sigma^2}\right)\right)$$

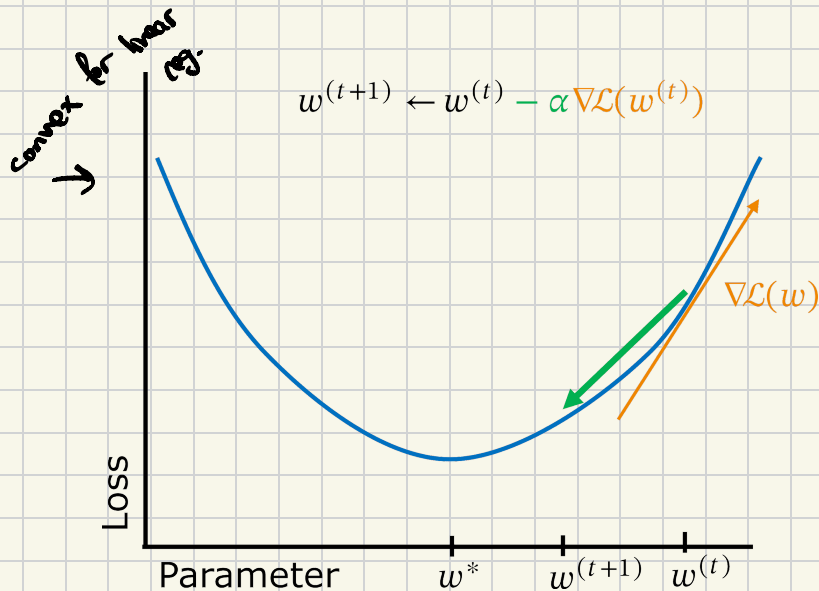
$$= \operatorname{argmin}_w -\sum_{i=1}^n \frac{(y^{(i)} - \langle w, x^{(i)} \rangle)^2}{2\sigma^2}$$

$$= \operatorname{argmin}_w \sum_{i=1}^n (y^{(i)} - \langle w, x^{(i)} \rangle)^2$$

Gradient Descent

Computing w^* for linear regression

takes $O(nd^2 + d^3)$ time, can we improve?



Initialize $w^{(0)}$ randomly

Choose learning rate α

$$w^{(t+1)} \leftarrow w^{(t)} - \alpha \nabla \mathcal{L}(w^{(t)})$$

Time to take T steps for linear regression?

Why Gradient descent?

1 Dimension $w \in \mathbb{R}$

$$\frac{\partial \mathcal{L}(w)}{\partial w} = \lim_{\Delta \rightarrow 0} \frac{\mathcal{L}(w+\Delta) - \mathcal{L}(w)}{\Delta}$$

For small Δ ,

$$\frac{\partial \mathcal{L}(w)}{\partial w} \approx \frac{\mathcal{L}(w+\Delta) - \mathcal{L}(w)}{\Delta}$$

$$\Leftrightarrow \mathcal{L}(w+\Delta) - \mathcal{L}(w) \approx \frac{\partial \mathcal{L}(w)}{\partial w} \Delta$$

We want $\mathcal{L}(w+\Delta) - \mathcal{L}(w) < 0$

$\Rightarrow$ choose $\Delta = -\alpha \frac{\partial \mathcal{L}(w)}{\partial w}$ so

$$\mathcal{L}(w+\Delta) - \mathcal{L}(w) \approx \frac{\partial \mathcal{L}(w)}{\partial w} \cdot -\frac{\partial \mathcal{L}(w)}{\partial w} \alpha < 0$$

$$w^{(t+1)} \leftarrow w^{(t)} + \Delta = w^{(t)} - \alpha \frac{\partial \mathcal{L}(w^{(t)})}{\partial w}$$

Higher Dimensions $w \in \mathbb{R}^d$

$$\frac{\partial \mathcal{L}(w)}{\partial w_i} = \lim_{\Delta \rightarrow 0} \frac{\mathcal{L}(w + \Delta e_i) - \mathcal{L}(w)}{\Delta}$$

$$\mathcal{L}(w + \Delta e_i) - \mathcal{L}(w) \approx \frac{\partial \mathcal{L}(w)}{\partial w_i} \Delta$$

For $v \in \mathbb{R}^d$

$$\mathcal{L}(w + \Delta v) - \mathcal{L}(w) \approx \sum_{i=1}^d \text{change along component } i$$

$$\approx \sum_{i=1}^d \frac{\partial \mathcal{L}(w)}{\partial w_i} \Delta v_i = \Delta \langle \nabla \mathcal{L}_w(w), v \rangle$$

$$\langle a, b \rangle = \|a\|_2 \|b\|_2 \cos(\theta)$$

choose $v = -\nabla_w \mathcal{L}(w)$ to align direction

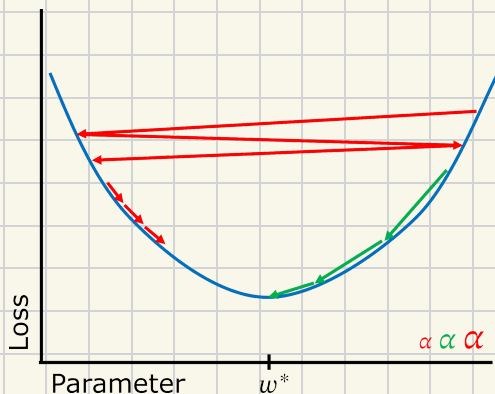
Stochastic Gradient Descent

Choose batch $S \subseteq \{1, \dots, n\} = [n]$

$$\mathcal{L}_S(w) = \frac{1}{|S|} \sum_{i \in S} (f(x^{(i)}) - y^{(i)})^2$$

$$w^{(t+1)} \leftarrow w^{(t)} - \alpha \nabla_w \mathcal{L}_S(w^{(t)})$$

Step Sizes



Choosing $\alpha \dots$

- Manual:

If loss decreases, try $\uparrow \alpha$ } by factors of 1/2
If loss jumps around, try $\downarrow \alpha$

- Decay:

Decrease as we "get closer"

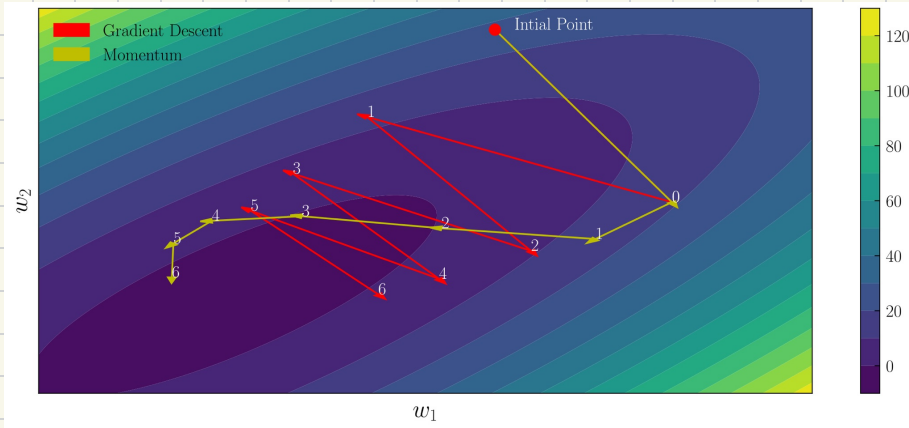
- Adaptive:

Choose proportional to inverse grad sum

$\rightarrow \uparrow$ grad, risk of overshooting

$\rightarrow$ use sum of past for stability

Momentum



$$v^{(t+1)} \leftarrow \beta v^{(t)} + \nabla_w \mathcal{L}_S(w^{(t)})$$

$$w^{(t+1)} \leftarrow w^{(t)} - \alpha v^{(t+1)}$$

$\beta \in [0, 1)$ controls "friction"